

Time-dependent Schroedinger equation (finite difference approach)

We use the 2D harmonic oscillator from the previous assignment

$$V(x, y) = \frac{1}{2}m(\omega_x^2 x^2 + \omega_y^2 y^2). \quad (1)$$

We assume a rectangular grid of size $N \times N$. The choice of N – as large as possible for a calculation that lasts not more than 2 minutes. We use the atomic units with $m = m_e = 1$, $\hbar = 1$. Lets take $\omega_x = 1$ and $\omega_y = 1.001$ (a change with respect to the last week).

We will solve the Schroedinger equation

$$i\hbar \frac{\partial \Psi}{\partial t} = H\Psi \quad (2)$$

for a time-independent potential.

The wave function in the initial moment is given by

$$\Psi(t=0) = \frac{1}{\sqrt{3}}(\Psi_0 + \Psi_1 + i\Psi_2), \quad (3)$$

Where Ψ_0, Ψ_1, Ψ_2 are the Hamiltonian eigenstates – the lowest-energy state, the first-, and the second-excited states, respectively.

Crank-Nicolson scheme

We use the Crank-Nicolson scheme for solution of the Schroedinger equation

$$\Psi(t + \Delta t) = \Psi(t) + \frac{\Delta t}{2i\hbar} (H\Psi(t) + H\Psi(t + \Delta t)) \quad (4)$$

Lets use an iterative scheme for the time-stepping

$$\Psi(t + \Delta t)^{\nu+1} = \Psi(t) + \frac{\Delta t}{2i\hbar} (H\Psi(t) + H\Psi(t + \Delta t)^\nu). \quad (5)$$

We can take $\Psi(t + \Delta t)^0 = \Psi(t)$.

The choice of the time-step and the number of iterations depend on our choice of N . For $\Delta x = \frac{1}{8}$ and $\Delta t = 0.001$ the number of iterations necessary for convergence is $\simeq 5$.

Tasks

- For our initial condition the motion of the packet should be periodic with period 2π .
- Lets cover the time over 2 nominal periods of motion.
- Plot the norm and the average energy as a function of time.
- Plot the average $\langle x \rangle(t)$ and $\langle y \rangle(t)$.

- Is $\langle y(\langle x \rangle) \rangle$ a closed loop?
- Can you make the wave packet turn in the other direction ?
- Can you make it oscillate in x or y direction without a rotation?
- What happens if $\Psi(0) = \Psi_i$ (the eigenstate is set as the initial state)?